

ARULMOZHIVARMAN SELVARAJ

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EDUCATION

Northeastern University — Master of Science in Computer Science (Khoury College) Sep 2026 – May 2028
Relevant Coursework: Database Management Systems (in progress)
Sri Krishna College of Engineering and Technology — B.Tech in Information Technology Oct 2022 – Mar 2026

TECHNICAL SKILLS

- **Languages:** Java, Python, SQL, C++, JavaScript (Node.js), HTML/CSS
- **Backend & Distributed Systems:** Spring Boot, Spring Security, Apache Kafka, Redis, AWS SQS, Microservices, Transactional Outbox, Idempotency, REST APIs
- **Cloud & DevOps:** AWS (Lambda, API Gateway, S3, EC2, IAM, SQS), Docker, Kubernetes (basic), CI/CD (GitHub Actions)
- **Databases & Testing:** MySQL, PostgreSQL, MongoDB, JUnit, concurrency testing
- **Core Concepts:** Data Structures & Algorithms, System Design, OOP, OS Concepts

TECHNICAL EXPERIENCE

Accenture — Software Engineering Intern | Bangalore, India June 2025

- Built a serverless employee onboarding workflow using AWS Lambda, API Gateway, and S3, reducing manual data entry by 80%.
- Automated backend data processing with SQS-triggered Lambda consumers, decoupling downstream integrations and reducing manual processing steps.

PROJECTS

Concurrent Double-Entry Payment Ledger | Java 17, Spring Boot, MySQL, React.js, Docker Live Demo

- Engineered a concurrent double-entry ledger using pessimistic row locking (SELECT FOR UPDATE) and JPA @Version optimistic locking, preventing race conditions and account overdrafts.
- Verified correctness with multithreaded concurrency tests simulating simultaneous transfers, ensuring balances never drift and every transaction's entries sum to zero.
- Enforced idempotent transfers via an Idempotency-Key header backed by a database unique constraint, and guaranteed reliable event delivery with an append-only ledger and the Transactional Outbox pattern.
- Deployed on Render with a React dashboard for account management, balance tracking, and transaction history.

Distributed Event-Driven Order Processing System | Spring Boot, Kafka, Redis, Docker

- Built event-driven microservices (Auth, Order, Inventory, Notification) communicating asynchronously over Kafka, with an independent database per service.
- Secured APIs with JWT authentication and added Redis caching; containerized with Docker Compose.

Health & Wellness Analytics Dashboard | React.js, Spring Boot, PostgreSQL, Docker

- Built Spring Boot REST APIs integrating the Edamam Nutrition API with server-side caching that reduced response latency by 40%, plus a React/Chart.js analytics dashboard.

Graph Visualizer Browser Extension | React.js, D3.js, WXT

- Built a browser extension for competitive programmers that renders interactive graph visualizations, using React and D3 on the WXT extension framework.

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Cloud Practitioner** (Udemy course certificate) and **Low-Level Design** (Udemy course certificate).
- **LeetCode Knight** — solved 1,000+ problems across graphs, dynamic programming, segment trees, tries, and string algorithms.